ការបង្កើតឯកសារប្រវត្តិសាស្ត្រខ្មែរ

Khmer History Documentary Builder

How to use the **khmer-history-doc** skill for Claude Code

Paper-cut 2D documentaries · Khmer narration · real maps · authentic evidence



Version 1.0 · October 2026 · github.com/chydevit/animinations-v3

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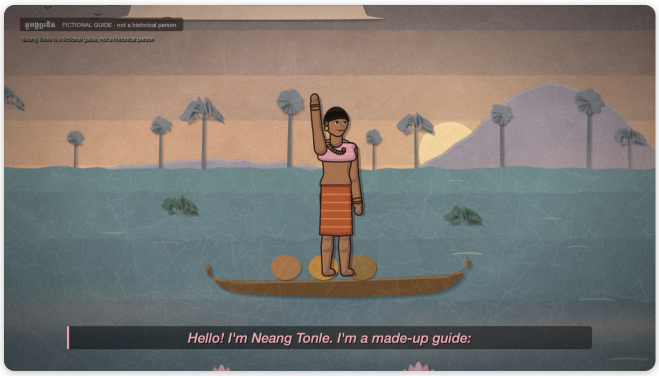
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> **In one sentence:** you give Claude a Cambodian history topic. It researches the facts, writes a Khmer and English script, draws paper-cut scenes, maps and evidence cards, records Khmer voices, mixes original music, and renders a finished, subtitled MP4 in which every claim and every picture is labelled honestly.

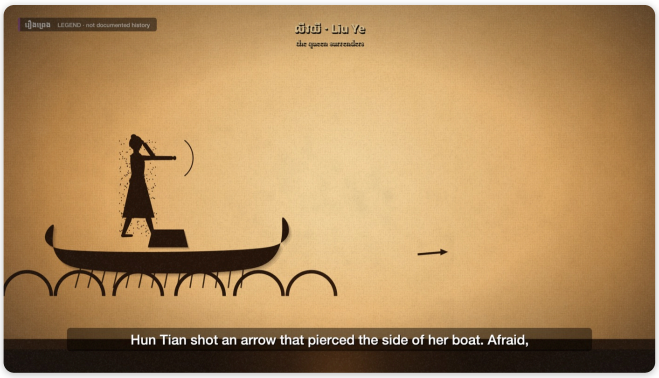
1. What the skill makes

This skill packages the documentary engine built over five films (Kampong Cham, Khmer Rouge, Preah Vihear, Ta Moan and Funan). Each film is a 10–20 minute educational documentary with:

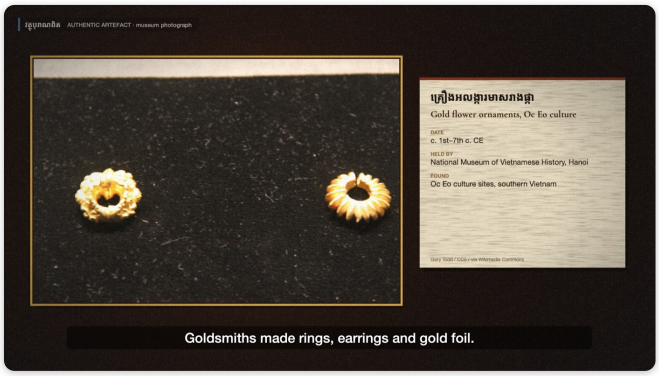
- **Paper-cut / Sbek shadow-puppet scenes**: layered paper sets with parallax and camera moves, film grain and transitions (ripple, lotus, ink).
- **A fictional child guide** (default: Neang Tonle). She asks a question at the start of each chapter. She is lip-synced to her own voice and always labelled fictional.
- **Animated maps** built from real elevation data and Natural Earth rivers, with markers, routes, moving ships and soft zones.
- **Authentic evidence**: Ken Burns photos, museum cards and classical Chinese text cards, each with licence and credit.
- **Khmer narration** in two locked VoxCPM2 voices (narrator and guide), checked with Khmer Whisper.
- **Original music** in pinpeat style (roneat, kong, khloy, tro, chapei), with ambience and SFX, mixed to –16 LUFS.
- **Subtitles**: English burned into the picture, plus soft English and Khmer tracks in the MP4.
- **Documents**: fact sheet, Khmer/English script, shot list, image manifest and ASR report.



The guide introduces herself (badge: FICTIONAL GUIDE).



A legend told as a shadow-puppet play (badge: LEGEND).



Museum card: a real object with its date, museum and licence.



Chinese dynastic record with a translation and its source.

2. Installation

You need: a Mac (Apple Silicon, 16 GB RAM or more recommended), Homebrew, Python 3.12, about 15 GB of free disk space, and Claude Code.

```
# 1. get the skill into Claude Code's skills folder
git clone https://github.com/chydevit/animinations-v3.git ~/.claude/skills/khmer-history-doc

# 2. one-time setup (tools, fonts, Python environments, VoxCPM2 model ~5 GB)
~/.claude/skills/khmer-history-doc/scripts/setup.sh
```

Setup step	What it installs
System tools	ffmpeg, cairo, pango (Homebrew)
Fonts	Noto Sans Khmer, Noto Serif Khmer, Moul, Cormorant Garamond
`.venv`	render environment: cairocffi, numpy, scipy, pillow, soundfile, pyloudnorm, pyshp
`.venv-vox`	VoxCPM2 + PyTorch (MPS) + transformers for the Khmer Whisper check. If `khmer-2d-animation` is installed, its environment is reused.

> To skip the voice environment, for example on a machine that only renders, run `SKIP_VOX=1 scripts/setup.sh`. Running setup again is always safe.

3. Quick start with Claude

Open Claude Code in your animation folder and describe the film. The skill starts automatically for any Khmer or Cambodian history documentary request. You can also type `/khmer-history-doc`.

```
cd ~/animation-2D
claude
> Make a 15-minute Khmer documentary about Sambor Prei Kuk and the Chenla kingdom,
  with Neang Tonle as the guide. Chapters: 0:45 the land, 3:00 Isanavarman I,
  6:00 the temples, 9:30 inscriptions, 12:30 legacy.
```

Claude then works through these stages:

1. Researches Khmer and English sources and writes `docs/fact_sheet.md`, with a tag on every claim.
2. Finds freely licensed images and records them in `archive/manifest.json`.
3. Creates the project, writes `narration.py` and `script_src.py`, and shows you preview sheets.
4. Designs voice candidates for you to choose from, then synthesises and checks every line.
5. Renders `film/<name>.mp4` plus a web copy, and reports the runtime and QC results.

Useful follow-up requests:

Say	What happens
“Show me the storyboard first”	Claude renders `make.py shotstills` and shows you the contact sheets
“Let me pick the voices”	Claude runs `make.py voices`, and you listen to `audio/voices/cand_*.wav`
“Line k3 sounds wrong”	Claude runs `make.py reseed k3` and then `make.py narrate`
“Make chapter 2 longer / add a map of the trade route”	Claude edits `script_src.py`, then previews before rendering
“Re-render only the changed part”	Claude deletes the affected chunks in `build/chunks/` and renders again

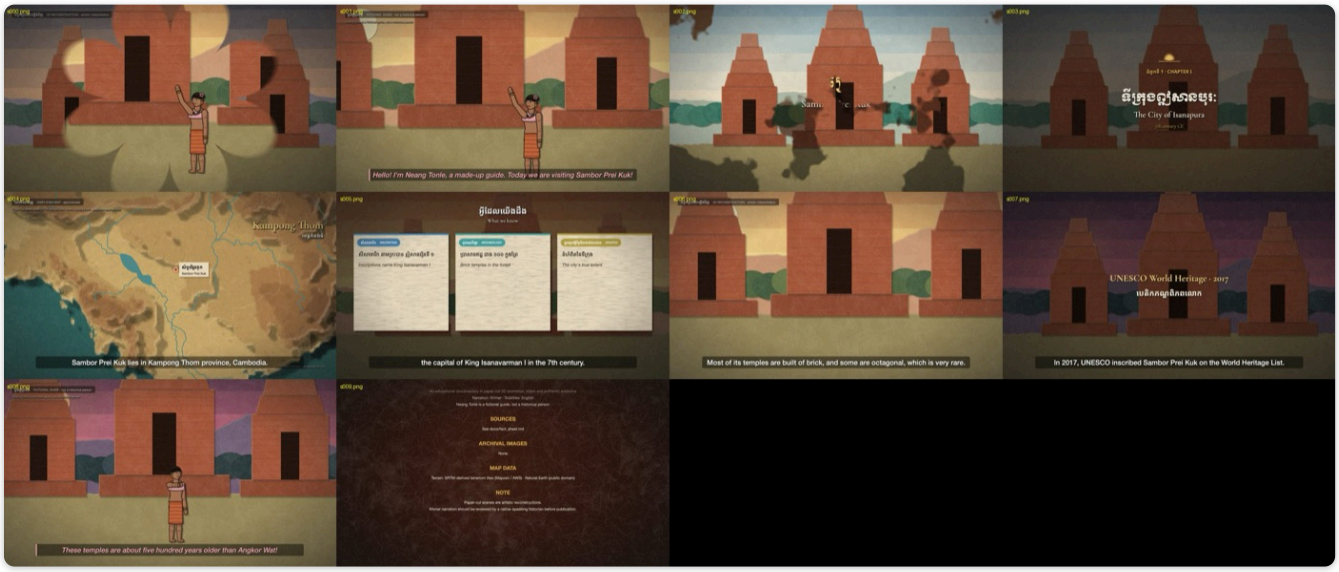
4. Quick start by hand

Every step can also be run without Claude. Each film is a self-contained folder with its own copy of the engine.

```
SK=~/.claude/skills/khmer-history-doc
$SK/.venv/bin/python $SK/scripts/new_doc.py ~/animation-2D/sambor_prei_kuk \
    --title-en "Sambor Prei Kuk" --title-km "សំបូរព្រៃគុក"
cd ~/animation-2D/sambor_prei_kuk

.venv/bin/python make.py info          # runtime and shot list
.venv/bin/python make.py shotstills    # build/shots/sheet_00.jpg
.venv/bin/python make.py voices        # 4 candidate voices per speaker
.venv/bin/python make.py lock nar 11   # choose the narrator
.venv/bin/python make.py lock tonle 23 # choose the guide
.venv/bin/python make.py narrate       # synthesise every line
.venv/bin/python make.py asr           # Khmer Whisper check
.venv/bin/python make.py all           # subtitles + audio + video -> film/sambor_prei_kuk.mp4
```

A new project contains a short working sample about Sambor Prei Kuk (about 70 seconds), so you can test the whole pipeline at once. To start from the complete 15-minute Funan film instead, add `--example funan`.



make.py shotstills on the sample project: one frame per shot, labelled with its badge and subtitle.

5. Inside a project

```
my_film/
  project.json      name, Khmer/English title, fps, end_target (total length in seconds, optional)
  narration.py      every spoken line (Khmer + English) and the VOICES of each speaker
  script_src.py     the shot list: what is on screen while each line is spoken
  make.py           the driver (all commands)
  engine/           private copy of the engine: patch freely for this film
  docs/             fact_sheet.md, script_km_en.md, shot_list.md, asr_report.md
  source/           saved source texts (Khmer and English)
  archive/          img/ + manifest.json  (authentic images with licences)
  geo/              Natural Earth shapes + elevation (dem_sea.npz, dem_delta.npz)
  audio/            voices/ (locked reference voices), narr/ (one wav per line), seeds.json
  build/            cache/, shots/, stills/, chunks/, mix.wav, logs
  subtitles/        <name>_en.srt, <name>_km.srt
  film/             <name>.mp4 (master), <name>_web.mp4, character_sheets/
```

File	You edit it to…
`narration.py`	change what is said, or the voice design of a speaker
`script_src.py`	change what is shown: sets, camera, maps, photos, cards, music, transitions
`project.json`	change the title, the output file name, or the target length
`engine/sets.py`, `cast.py`	add new places, buildings or people for a new era
`archive/manifest.json`	register authentic images (licence, credit, museum, date)

6. Step-by-step workflow

#	Step	Command / file	Check
1	Research the facts	`source/`, `docs/fact_sheet.md`	every claim tagged; DO-NOT-SAY list
2	Collect images	`archive/img`, `manifest.json`	free licence, correct caption
3	Create the project	`new_doc.py`	`make.py info` runs
4	Write the narration	`narration.py`	short sentences, Khmer digits
5	Build the shot list	`script_src.py`	cold open, chapters, badges, credits
6	Preview	`make.py shotstills`	look at every sheet
7	Voices	`voices` → `lock` → `narrate`	listen to candidates
8	Quality check	`make.py asr`	mean ≥ 0.85; reseed lines < 0.78
9	Render	`make.py all`, `make.py web`	watch the whole film
10	Hand-off	`make.py docs`	native-speaker review of the Khmer

Preview before voices. Until the voice clips exist, line lengths are estimated from the text, so `info` and `shotstills` work from the first minute. After narration the timeline adjusts itself to the real clip lengths.

Render time. About 0.35 s per frame on 6 workers, so a 15-minute film takes about 40–60 minutes. Voices take about 35–120 s per line, roughly 2–3 hours for 120 lines.

7. Writing the narration

`narration.py` holds every spoken line. `L()` is the narrator and `T()` is the guide. The key (for example `a2`) links the line to a shot in `script_src.py`.

```
VOICES = {
    "nar":   dict(label_en="", label_km="",
                  design="A warm, calm Cambodian man in his forties, documentary narrator ...",
                  sample="នៅសតវត្សទីប្រាំពីរ ទឹកដីកម្ពុជា មានរាជធានីដ៏ធំមួយ ..."),
    "tonle": dict(label_en="NEANG TONLE: ", label_km="នាងទន្លេ៖ ",
                  design="A bright, curious Cambodian girl about twelve years old ...",
                  sample="សួស្តី! ខ្ញុំឈ្មោះ នាងទន្លេ។ ..."),
}

T("t0", "សួស្តី! ខ្ញុំឈ្មោះ នាងទន្លេ។ ខ្ញុំជាតួអង្គប្រឌិត។ ថ្ងៃនេះ យើងទៅលេង សំបូរព្រៃគុក!",
  "Hello! I'm Neang Tonle, a made-up guide. Today we are visiting Sambor Prei Kuk!")
L("a2", "អ្នកប្រវត្តិវិទូ យល់ថា ទីនេះ គឺជាទីក្រុង ឥសានបុរៈ ... នៅសតវត្សរ៍ទី ៧។",
  "Historians identify it as Isanapura, the capital of King Isanavarman I in the 7th century.")
```

Writing tips

- One idea per line, and about 10–25 Khmer words. Spaces between clauses become natural pauses in the voice.
- Write numbers with Khmer digits (๖๑๑). `khnum.speak()` turns them into words for the voice.
- Put the type of evidence into the sentence: “A legend tells…”, “Chinese records say…”, “Historians still debate…”.
- Begin each chapter with a question from the guide, and let the narrator answer it.
- Editing a line re-synthesises only that line, because clips are keyed by a hash of the text, the seed and the speaker.

8. Building the shot list

`script_src.py` is a list of shots in order. Each shot shows a picture while its lines are spoken, so the length of the shot comes from the voice clips. Edit only the SHOTS section; the helpers and the timeline assembly stay the same.

```
CH(1, "ទីក្រុងឥសានបុរៈ", "The City of Isanapura", "7th century CE",
   under=lambda: X.make_chenla("day"), music="kings", amb="forest")

GUIDE("t0", lambda m: X.make_chenla("golden", people=[grp(0.6, guide(1500, 1150, 330, "wave"))],
                                    mouth_fn=m),
      cam((1450, 900, 1.5), (1450, 880, 1.6)), note=GUIDE_NOTE, trans="lotus")

S(["a3"], lambda: X.make_chenla("golden"), cam((1250, 900, 1.3), (1250, 860, 1.6)), badge="recon")

S(["a2"], lambda: cards.claim_board("អ្វីដែលយើងដឹង", "What we know", [
    ("inscription", "សិលាចារឹក ...", "Inscriptions name King Isanavarman I"),
    ("debated", "ទំហំពិតនៃទីក្រុង", "The city's true extent")]))
```

Helper	Shows
`S(lines, vis, cam, badge, …)`	any scene: a set, a card or a custom drawing
`GUIDE(key, scene_fn, cam)`	the guide speaking, with lip-sync from her clip
`CH(num, km, en, years, …)`	chapter card over the chapter's set
`M(lines, view_a, view_b, elements)`	animated map (see section 9)
`PH(file, lines)` / `MC(file, lines, …)`	authentic photo / museum card
`ZH(lines, zh, en, source)`	classical Chinese text card



Chapter card (CH).



Claim board: what we know and what is debated.

Camera, timing and sound options

Option	Meaning
`cam((x, y, zoom), (x, y, zoom))`	move across the 2400×1350 set; zoom 1.0 shows 1920 px of it
`pre` / `post` / `min_d`	seconds before the first line / after the last line / minimum length
`xin`, `trans`	overlap with the previous shot; transition: fade, cut, ripple, lotus, ink
`fade_in`, `fade_out`	fade from or to black
`music`	starts a mood: opening, land, legend, port, water, fleet, envoys, india, life, kings, decline, legacy, credits
`amb`	ambience: delta, river, forest, market, port, sea, lamp, workshop, temple, village, night, none
`sfx`	extra sounds, for example `("hit",)` or `("bell",)`
`TARGETS`, `end_target`	chapter start times and total length: holds stretch to meet them, and narration is never cut

Sets and people

Sets: `make_delta`, `make_lowland`, `make_puppet`, `make_port`, `make_sea`, `make_palace`, `make_temple`, `make_workshop`, `make_monastery`, `make_chenla`. Each one takes a time of day: dawn, day, golden, dusk, night or lamp. Figures are placed with `F(x, y, height, who, pose, yaw)` inside `grp(depth, …)`.



The built-in cast (build/character_sheet.py). Add new people for other eras in engine/cast.py.

9. Maps

Maps are drawn as layers of paper terrain cut from real elevation data, with Natural Earth rivers and lakes. Modern borders appear only faintly. A view is `(longitude, latitude, width in degrees)`, and the camera glides from `view_a` to `view_b`.

```
M(["a1"], (104.9, 12.5, 9.0), (104.9, 12.5, 7.5), [
    dict(k="marker", lon=105.043, lat=12.871, km="សំបូរព្រៃគុក", en="Sambor Prei Kuk", at=1.0,
pulse=True),
    dict(k="blob", pts=[(104.6, 12.4), (105.6, 12.3), (106.1, 15.3)], fill="#d8aa4a", at=2.0),
    dict(k="line", pts=[(104.97, 10.95), (105.04, 12.87)], at=1.5, ships=2),
    dict(k="note", km="...", en="Extent approximate", cls="debated", at=3.0),
    dict(k="era", text="7th century", km="សតវត្សរ៍ទី ៧", at=0.5)])
```



Map shot from the sample project: paper terrain, a pulsing marker and an era label.

> The default map data covers mainland Southeast Asia (wide view) and Cambodia plus the Mekong delta (close view, used when the view is narrower than 9°). For another region, run `.venv/bin/python ~/.claude/skills/khmer-history-doc/scripts/fetch_geo.py . --local 99 12 106 18 8`

10. Evidence: photos and records

Use only freely licensed images (public domain, CC0, CC BY, CC BY-SA), checked on the source page. Put the file in `archive/img/` and add an entry to `archive/manifest.json`:

```
{ "file": "img/spk_prasat_yeai_poeun.jpg",
  "object_name": "Prasat Yeay Poan, Sambor Prei Kuk",  "object_name_km": "ប្រាសាទយាយពៅន",
  "object_date": "7th century CE (photo 2015)",      "holding_institution": "in situ",
  "licence": "CC BY-SA 4.0",
  "attribution_line": "Author / CC BY-SA 4.0 / via Wikimedia Commons",
  "authenticity_note": "what the image really shows, and what it does not" }
```

```
PH("spk_prasat_yeai_poeun.jpg", ["a3"])                    # photo with a slow Ken Burns move
MC("spk_lintel.jpg", ["a5"], date="7th c.", inst="National Museum of Cambodia")   # museum card
ZH(["k2"], "扶南國...", "“The kingdom of Funan ...”", "Liang shu 54 (636 CE)")   # Chinese record
```

The credits list every image that was used, with its licence. If an image file is missing, `PH` falls back to a text card, so a missing download never breaks the render.

11. Voices and quality check

Choose and lock the voices

```
.venv/bin/python make.py voices          # audio/voices/cand_nar_11.wav … cand_tonle_51.wav
open audio/voices                        # listen
.venv/bin/python make.py lock nar 23     # copies cand_nar_23.wav -> nar.wav
.venv/bin/python make.py lock tonle 37
```

Each speaker is cloned from one locked reference clip, so the voice stays the same through the whole film, and across films if you copy `audio/voices/`.

Synthesise

```
.venv/bin/python make.py narrate     # resumable; 20 lines per fresh worker; keeps the Mac awake
```

Check with Khmer Whisper

```
.venv/bin/python make.py asr
#  mean 0.912 low [('k3', 0.71)]
#  fix: make.py reseed k3 && make.py narrate
```

`asr` transcribes every clip with a Khmer Whisper model and compares the result with the script. The full table is in `docs/asr_report.md`. A low score usually means a skipped or garbled word, so re-seed the line and synthesise it again.

> **Native review is still required.** The ASR check catches pronunciation failures, but not awkward or wrong Khmer. Have a native-speaking historian read `docs/script_km_en.md` before you publish.

12. Rendering and outputs

```
WORKERS=6 .venv/bin/python make.py all   # subs + audio mix + video + mux
.venv/bin/python make.py web             # smaller H.264 copy for YouTube / Facebook
.venv/bin/python make.py docs            # docs/script_km_en.md, docs/shot_list.md
.venv/bin/python build/character_sheet.py tonle   # character sheets
```

Output	Contents
`film/<name>.mp4`	1920×1080, 24 fps, burned English subtitles, AAC 256k, soft en + km subtitle tracks
`film/<name>_web.mp4`	same film at a smaller size (CRF 24)
`subtitles/*.srt`	English and Khmer subtitle files
`build/mix.wav`	the final mix: voices, music, ambience and SFX at −16 LUFS
`docs/`	fact sheet, script, shot list, ASR report

The video is rendered in 10-second chunks (`build/chunks/`) in parallel. If the render stops, run the same command again and it continues where it stopped. After a visual change, delete the affected chunks (or all of them) before rendering again.

You can change one part of the film without starting over:

You changed…	Run
a narration line	`make.py narrate` → `make.py all` (delete the chunks after that point)
a picture only	delete the chunks for that shot → `make.py render`
music or ambience	`make.py audio` → `make.py mux`
subtitle text	`make.py subs` → delete the chunks (subtitles are burned in) → `make.py render`

13. Accuracy rules and labels

The engine is built around one rule: **accuracy beats drama**. The audience must always know whether they are looking at evidence or at an artist's reconstruction.

Fact-sheet tag	How it is narrated
DOCUMENTED (source, year)	“According to the Book of Liang (636)…”
INSCRIPTION (K. number)	“An inscription carved in stone says…”
ARCHAEOLOGY	“Archaeologists have found…”
LEGEND	“A legend tells…”
DEBATED	“Historians still debate…” (give both sides)
HYPOTHESIS	“Some historians think…”
UNVERIFIED	not narrated

On-screen badge	Use for
recon	paper-cut reconstructions (artistic interpretation)
legend	legends and origin stories
guide	every shot of the fictional guide
photo / artifact / inscription / satellite	authentic images, always with a credit line
text	historical records (translated)
map / map_recon / schematic	maps: approximate positions or reconstructed extents
comparison	an object from another place or period, shown for comparison

- Ancient kingdoms are drawn as soft zones, never with hard borders.
- Contested topics are given with each side's claim and its source, in a neutral tone.
- Every fact sheet ends with a **DO NOT SAY** list of popular myths found during research.



Every reconstruction carries its badge in the top-left corner.

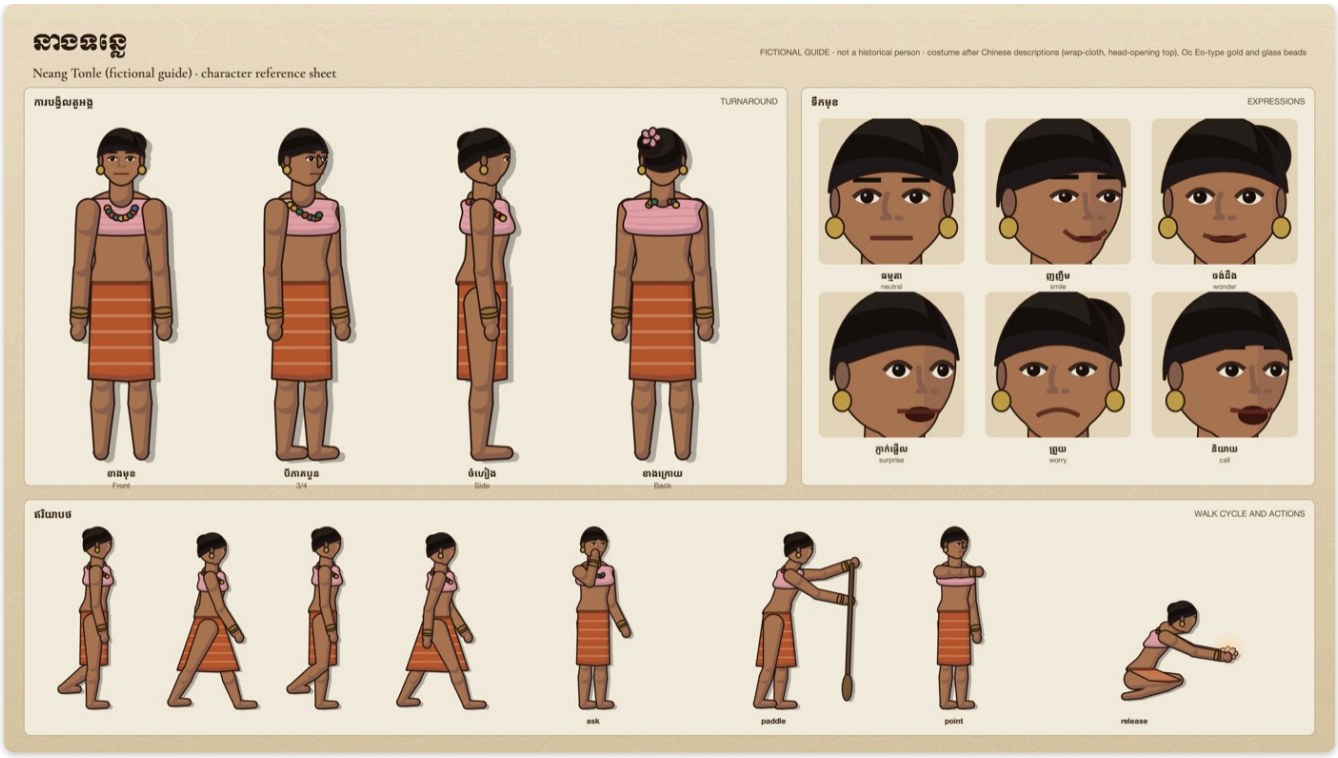
14. Customising the look

Every project has its own copy of `engine/`, so you can change it freely without affecting other films.

To add…	Edit	How
a new place	`engine/sets.py`	write `make_myplace(tod, people, mouth_fn)` that returns paper layers (see make_chenla)
a new person or costume	`engine/cast.py`	add a `CAST` entry: body, skin, hair, wrap, top, ornaments
a new music mood	`engine/audio.py`	add to `MOTIFS`: root note, scale, cell, lead instrument, tempo
a new badge	`script_src.py`	`cards.BADGES['key'] = (colour, km, en)`
colours	`engine/core.py`	`PAL` palette (terracotta, ochre, lotus, jade, gold…)
a different guide	`narration.py` + `cast.py`	new VOICES entry and CAST entry; set `GUIDE`

```
def make_myplace(tod="golden", people=None, mouth_fn=None):
    P = TOD[tod]
    def bg(ctx):
        PP.sky(ctx, P, 0.55, (0.3, 0.35), 70, 7, 140)
        PP.ridge(ctx, P, 720, 90, mix(P["far"], P["haze"], 0.3), 141)
    def ground(ctx):
        sheet(ctx, path_poly(jag([(-30, 940), (SW + 30, 940)], 3, 160) + [(SW + 30, SH + 30), (-30,
SH + 30)]),
              "#9a8a5a", shadow=0.6, seed=160)
        PP.palm(ctx, P, 400, 1000, 600, "sugar", 3, P["near"])
    layers = [(_key("mp_bg", tod=tod), bg, 0.25), (_key("mp_g", tod=tod), ground, 0.6)]
    return dict(layers=layers, dyn=people_dyn(people or [], P, mouth_fn), dyn_depths=[0.25, 0.6])
```

Layers are cached as PNG files by key. After you change a painter, delete `build/cache/layers/`. Check new pictures with `make.py stills 12.0` before rendering.



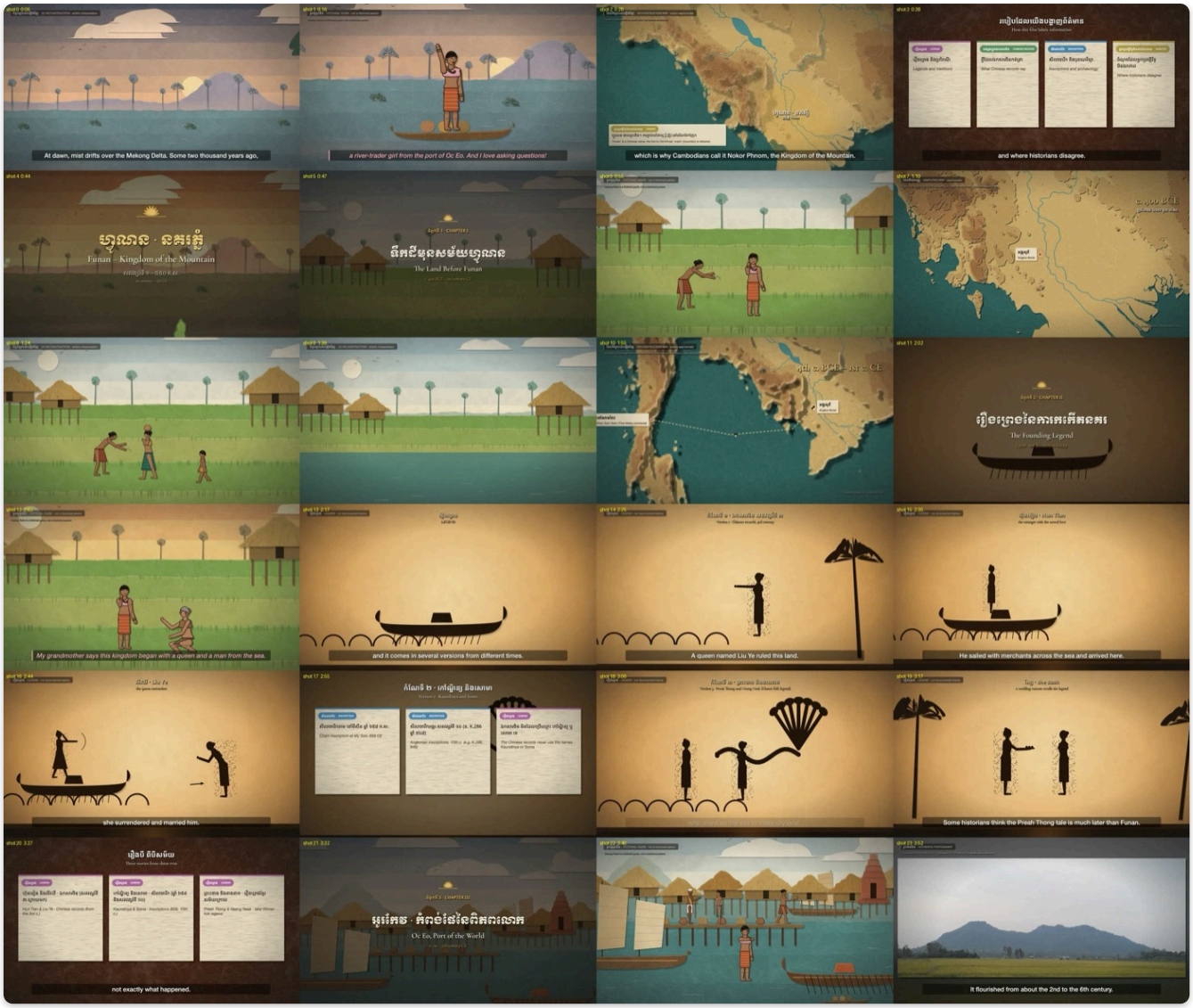
Character sheet for the guide: turnaround, expressions, walk cycle and actions.

15. Troubleshooting

Problem	Fix
Khmer shows as boxes	fonts are missing: run `fc-list | grep Khmer`, then run setup.sh again
“VoxCPM2 environment not found”	run setup.sh, or set `KHDOC_VOX_PY=/path/to/python`
“no locked voice for ['tonle']”	run `make.py voices`, then `make.py lock tonle <seed>`
Voice synthesis gets very slow	normal for VoxCPM2 on MPS; `narrate` restarts the worker every 20 lines. Close other apps; never run two workers
A line skips or garbles words	`make.py reseed <key>` → `make.py narrate`
The credits are cut off at the end	never add `-shortest` to ffmpeg when subtitle tracks are present
An old picture still appears after an edit	delete `build/chunks/` (and `build/cache/layers/` if a painter changed)
A punched hole cuts through the whole frame	draw Cairo OPERATOR_CLEAR inside `push_group()`
A map is blank or flat	run `scripts/fetch_geo.py <project>`; check `geo/dem_*.npz`
Chapters start at the wrong times	set `TARGETS` (one per chapter) and `end_target`; holds stretch but never shrink

16. Command cheat sheet

Command	Does
`scripts/setup.sh`	one-time install
`scripts/new_doc.py <dir> --title-en … --title-km …`	new project (add `--example funan` for the full example)
`scripts/fetch_geo.py <dir> [--local …]`	map data
`make.py info`	runtime, chapter times, shot list
`make.py shotstills [i …]`	preview frames + contact sheets
`make.py stills t …`	frames at given times
`make.py voices` / `lock who seed`	design / lock voices
`make.py narrate`	synthesise missing lines
`make.py asr` / `reseed key …`	QC / redo lines
`make.py subs` / `audio`	subtitles / music + voice mix
`make.py render` / `video` / `mux`	full render / picture only / final mux
`make.py all`	subs + audio + render
`make.py web` / `docs`	web copy / script and shot-list docs
`build/character_sheet.py [who]`	character sheets



Storyboard sheet from the Funan film (15:41), the full example in examples/funan/.

Map data: Natural Earth (public domain) and SRTM-derived terrarium tiles (Mapzen / AWS Open Data). Voices: VoxCPM2 (OpenBMB). ASR check: seanghay/whisper-small-khmer-v2. Archival images in the screenshots keep their own licences, as credited on screen.